

University of Yaoundé I
Department of Physics, Faculty of Science
P.O. Box 812, Yaoundé, Cameroon
myke-vital.sao@facsciences-uy1.cm
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Dear Editors,
13 September 2026

We submit the manuscript entitled “**Deconstructing Out-of-Distribution Extrapolation in Molecular Deep Learning: Structural Complexity, Topological Rescue, and Conformal Triage on African Antimalarial Natural Product Scaffolds**” for consideration as a Research Article in the *Journal of Computer-Aided Molecular Design*.

Rather than offering another passive benchmark reporting fingerprint dominance, this study delivers a mechanistic explanation for graph neural network extrapolation failures and an actionable decision framework for computational virtual screening under out-of-distribution (OOD) shift. On a frozen panel of 19 836 African antimalarial natural products and synthetic derivatives, evaluated across 25 fold–seed replicates per model, we establish three major contributions:

First, we demonstrate that standard GNN performance drops sharply on stereochemically complex 3D polycyclic scaffolds ($Fsp^3 \geq 0.45$, ring count ≥ 4 , ROC AUC dropping to 0.768), exposing the theoretical 1-WL Weisfeiler–Lehman graph expressivity bottleneck. Fusing persistent-homology topological feature projections (GIN–TFP) injects global structural shape invariants (H_0, H_1), recovering up to 36 % of the extrapolation gap (0.815 ROC AUC, a +0.047 gain over base GIN).

Second, we convert these distance-resolved insights into an actionable Distance-Aware Conformal Triage Filter. By dynamically routing low-similarity candidates ($D_{NN} < 0.40$) to ECFP4–RF and mid-range candidates to GIN–TFP while monitoring calibration degradation ($ECE > 0.08$), this triage protocol boosts prospective virtual screening precision by +14.2 % on a 22 267-molecule disjoint external ChEMBL transfer panel.

Third, we establish a normative benchmark standard by proving that fold-independent weight restoration in molecular sequence transformers is mandatory to prevent artificial cross-fold information leakage during cross-validation.

The work is original, is not under consideration elsewhere, has been approved by all authors, and the authors declare no competing financial interests.

Suggested International Reviewers:

1. **Prof. Kelin Xia** (Nanyang Technological University, Singapore) — Expert in topological deep learning, persistent homology, and molecular graph representations. Email: xiakelin@ntu.edu.sg
2. **Prof. Fidele Ntie-Kang** (University of Buea, Cameroon / Martin Luther University Halle-Wittenberg, Germany) — Lead architect of the African Natural Products Database (ANPDB) and expert in natural product cheminformatics. Email: fidele.ntie-kang@uni-halle.de
3. **Prof. Carsten Eickhoff** (University of Tübingen, Germany) — Expert in molecular representation benchmarking and out-of-distribution generalisation. Email: c.eickhoff@uni-tuebingen.de

We thank you for your consideration.

Myke Vital Sao Temgoua
Corresponding Author